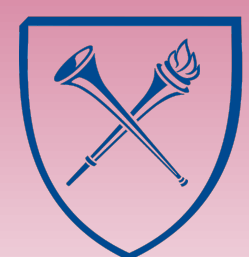


THE KNESER GRAPH OF TRIANGULATIONS

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Abstract

The Kneser graph of the set of k -element subsets of $[n] := \{1, \dots, n\}$ has been a central object of study in extremal set theory. Here, we present recent results on the Kneser graph of the set of triangulations of a convex n -gon. Joint works with Anton Molnar, Cosmin Pohoata, and Daniel G. Zhu.

The Kneser graph $\text{KG}(\mathcal{T}_n)$

Let $\mathcal{F}$ be a family over some (finite) ground set X . We define $\text{KG}(\mathcal{F})$ to be the graph with

- vertex set $\mathcal{F}$, and
- edges by $F \sim F' \iff F \cap F' = \emptyset$.

For our setting, we identify triangulations as $(n-3)$ -subsets of its diagonals. Labeling the vertices $1, \dots, n$, the family of triangulations, denoted $\mathcal{T}_n$, thus has ground set

$$\text{Diag}_n := \left\{ ij \in \binom{[n]}{2} : i - j \not\equiv_n \pm 1 \right\}.$$

Example 1 For $n = 5$, we get $\text{KG}(\mathcal{T}_5) \simeq C_5$.

Note that, in general, $\text{KG}(\mathcal{T}_n)$ is *not* vertex-transitive.

Fact 2 (Goldbach) $\mathcal{T}_n = C_{n-2} \in \Theta(4^n/n^{3/2})$ where C_n is the n th Catalan number.

The chromatic number of $\text{KG}(\mathcal{T}_n)$

In a highly influential paper from 1978, Lovász used topological methods to resolve Kneser's conjecture from 1955.

Theorem 3 (Lovász, 1978) If $n \geq 2k$, then

$$\chi(\text{KG}([n]^{(k)})) = n - 2k + 2.$$

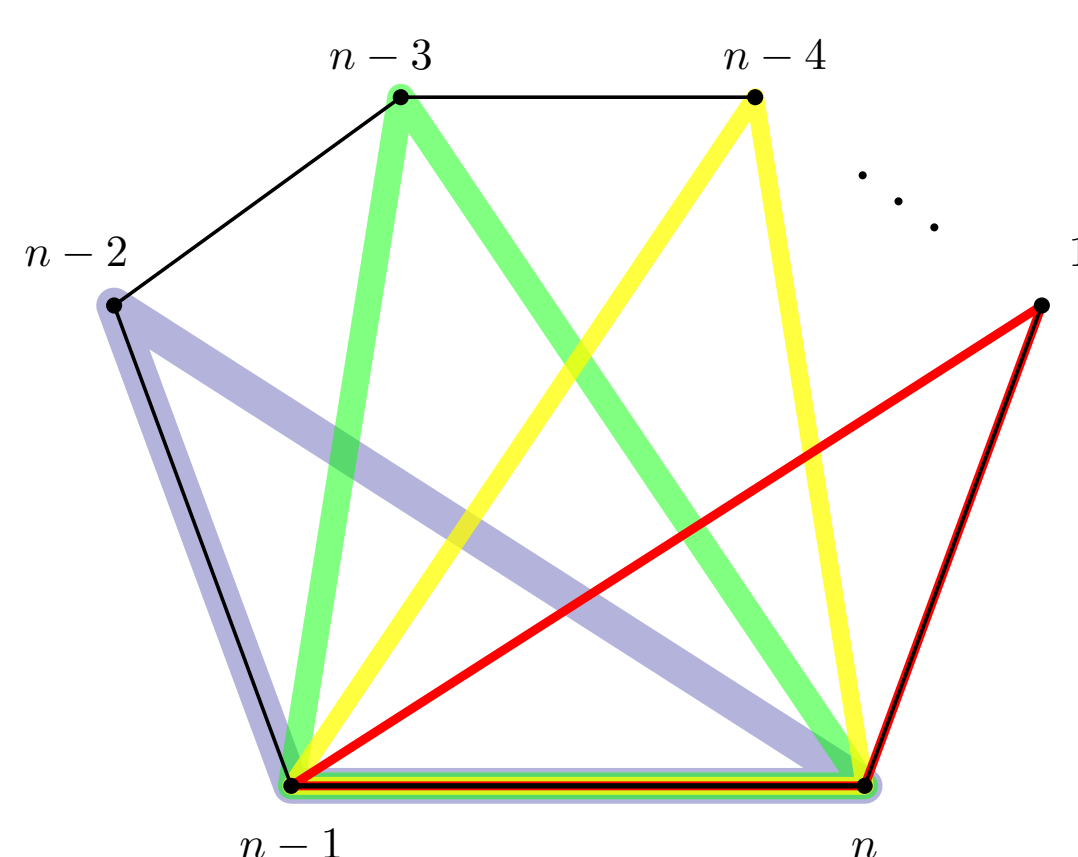
We prove an analogous result for $\text{KG}(\mathcal{T}_n)$ and some strengthenings paralleling the literature.

Theorem 4 (MPZZ, 2025+) For $n \geq 4$, we have

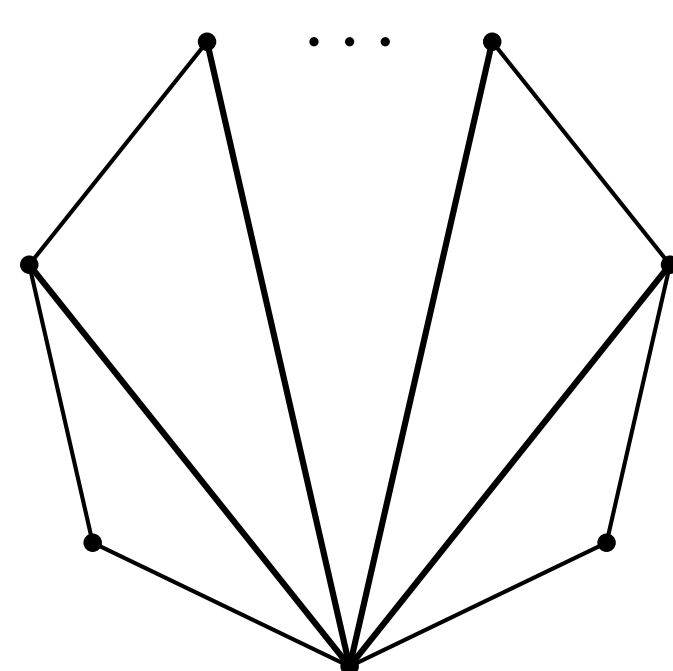
- $\chi(\text{KG}(\mathcal{T}_n)) = n - 2$.
- $\chi(\text{KG}(\mathcal{T}_n - T)) = n - 3 \iff T$ is a star. (“stability”)
- There exists a subfamily $\mathcal{T}_n^{(3)} \subseteq \mathcal{T}_n$ of size

$$|\mathcal{T}_n^{(3)}| = F_{2n-5} \in \Theta(2.618 \dots^n)$$

such that $\chi(\text{KG}(\mathcal{T}_n^{(3)})) = n - 2$. (“towards Schrijver”)



$$\chi(\text{KG}(\mathcal{T}_n)) \leq n - 2$$

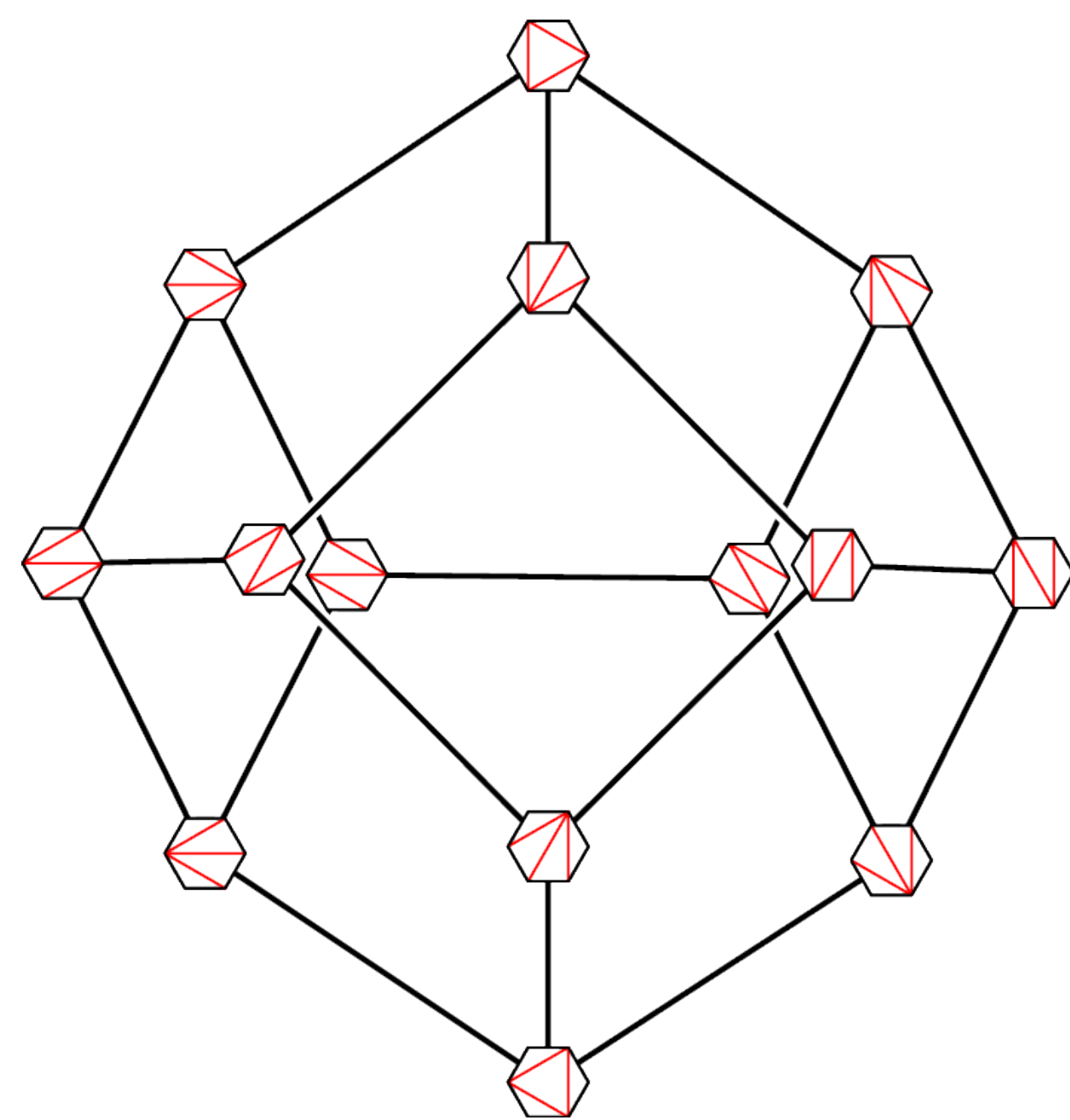


A star triangulation

The associahedron

A key ingredient for Theorem 4 is the *associahedron* Assoc_{n-3} , an $(n-3)$ -dimensional combinatorial polytope whose

- vertices v_T correspond to triangulations T , and
- facets f_d correspond to diagonals d , such that
- v_T is incident to f_d if and only if $d \in T$.



Assoc_3 (Picture by John C. Baez)

To show Theorem 4, we use the *Gelfand–Kapranov–Zelevinsky realization* of Assoc_{n-3} paired with the Lyusternik–Schnirelmann theorem, a consequence of the Borsuk–Ulam theorem.

The Hamiltonicity of $\text{KG}(\mathcal{T}_n)$

As a special case of Lovász conjecture, the Hamiltonicity of Kneser graphs has been of great interest. This was resolved in

Theorem 5 (Merino–Mütze–Namrata, 2025) Let $n \geq 2k + 1$, $(n, k) \neq (5, 2)$. Then $\text{KG}([n]^{(k)})$ is Hamiltonian.

We show that the same is true for $\text{KG}(\mathcal{T}_n)$.

Theorem 6 (MPZ, 2026+) $\text{KG}(\mathcal{T}_n)$ is Hamiltonian for $n \geq 5$.

The proof is combinatorial and makes use of the following

Observation 7 Acting on $\mathcal{T}_n$ by clockwise rotation, the orbits form cycles in $\text{KG}(\mathcal{T}_n)$.

We “glue” these cycles by using a cycle in the *flip graph* of $\mathcal{T}_n$. For this, the fact that the flip graph itself is Hamiltonian is crucial. This was shown in

Theorem 8 (Lucas, 1987) Let $n \geq 5$. There exists an enumeration $T_1, \dots, T_M$ of $\mathcal{T}_n$ such that $|T_1 \triangle T_2| = |T_2 \triangle T_3| = \dots = |T_{M-1} \triangle T_M| = |T_M \triangle T_1| = 2$.

Open problems

Analogous to the celebrated Erdős–Ko–Rado theorem, the following conjecture was independently posed by Kalai and Meagher.

Conjecture 9 (Kalai–Meagher, 2012) $\alpha(\text{KG}(\mathcal{T}_n)) = C_{n-3}$.

Note that this would be tight as witnessed by the family of triangulations containing a fixed short diagonal.